

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st May 2014

**Subject CA1 – Actuarial Risk Management
(Paper II)**

Time allowed: 3 Hours (14.45* - 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Attempt all questions, beginning your answer to each question on a separate sheet.*
3. **You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have 3 hours to complete the paper.*
4. *You must not start writing your answers until instructed to do so by the Supervisor.*
5. *Mark allocations are shown in brackets.*
6. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Explain why an investment portfolio may be valued using a notional portfolio rather than the actual portfolio? [5]
- Q. 2)** A general insurance company writes all major classes of business. Discuss the consequences of underestimating claim inflation assumption while determining claims reserves. [5]
- Q. 3)**
- i)** State under what economic conditions would money market instruments be attractive investments? (4)
 - ii)** Life insurers normally have long term liabilities. Give reasons, why a life insurance company would invest in money market instruments. (4)
- [8]
- Q. 4)** A life insurance company while pricing a whole of life immediate annuity contract assumed an interest rate of 1% p.a. higher than the gross redemption yield on an appropriate government security. The single premiums received were invested as follows:
- | | |
|-----------------|--------|
| Corporate bonds | - 75% |
| Equity | - 25%. |
- i)** Discuss the appropriateness of the above investment policy (7)
 - ii)** Suggest a possible alternative investment strategy that the company could have considered. (3)
- [10]
- Q. 5)** A general insurance company (Company G) has a large market share in property insurance business. Company L is a wholly owned subsidiary of Company G, and is a small life insurance business specializing in without profit term insurance covering mortgages.
- Company X is a medium sized life insurance company writing all lines of business (with and without profit). It wants to make a bid to buy Company L.
- Discuss the financial and operational issues that Company X should consider before it makes an offer. [12]
- Q. 6)**
- i)** Describe the factors that need to be considered while setting the discontinuance terms for a life insurance product? (6)
- A medium sized life insurer has been receiving a lot of surrender requests in the last 6 months on its traditional with profit portfolio. The CEO is worried and feels that efforts must be made to contain this sudden increase in surrenders. The Sales Director highlights that the company is paying less than the asset share to these policyholders and hence making surrender profits.
- ii)** Explain why you think there could be a sudden increase in surrenders. (4)
 - iii)** What issues the company needs to consider before taking a decision on whether or not to take any action to contain surrenders. (3)
- [13]

- Q. 7)** In a country, the insurance regulator is reviewing solvency norms for insurance companies. The regulator is considering allowing companies to use their own internal capital models.

Your company sells primarily risk products and unit linked products (with and without guarantees). It has asked you to prepare a report highlighting principal risks that need to be considered when building an internal capital model.

Discuss the points that you would cover in your report.

[14]

- Q. 8)** A very well-known wedding planner has approached a general insurance company to introduce a product covering all risks that his customers are exposed to during a wedding. The product will be marketed by the wedding planner only.

- i)** State reasons, why the insurer should sell this product only through the wedding planner (4)
- ii)** List the main risks the policy can cover (5)
- iii)** State the rating factors the insurer can use in setting the premium rates (5)

[14]

- Q. 9)** **i)** State the factors that may lead to contribution risks for both defined benefit and defined contribution schemes. (4)

- ii)** State why a defined benefit scheme would analyse its surplus? (5)

- iii)** The triennial funding valuation of a defined benefit scheme has just been completed. The following information is available to you.

- Surplus arising during the inter valuation period is INR 600 Million. An analysis of surplus reveals the following-
 - One off surplus from equity sales- INR 500 Million
 - Surplus from other sources-INR 100 Million

Two possible options are available to deal with the surplus arising.

Option1 - The employer's contribution rate can be reduced from 10% to 5% of pensionable salaries for each of the next three years.

Option 2 - The rate of increase of pensions in payment could be changed from a fixed 4% per annum to 5% per annum. This change to the pension increase would apply to pensioners and deferred pensioners as well as current contributing members.

These two options have the same expected cost.

Discuss the 2 proposals for using the surplus.

(10)

[19]
